Information Decomposition in Marathon Finish-Time Prediction

Harris Bubalo[1], Indrajeet Roy[1], Jacob Waga[1], Malick Tobe[1], Scott Bradley[1]

[1]Northeastern University

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***Abstract*—We study where predictive accuracy comes from in marathon finish-time prediction. We decompose accuracy along a runner's information timeline, from registration to in-race checkpoints, and measure how much each information source contributes. A nested five-model ablation with bias-corrected and accelerated cluster-bootstrap confidence intervals quantifies the marginal gain from demographics, prior race history, and cumulative split times. Conformal prediction provides distribution-free uncertainty bands that shrink as in-race data accumulates, marking the point at which in-race models surpass the best pre-race model in calibrated precision.**

Using Boston Marathon data from 2000–2019 (347,188 training observations; 26,214 test observations from 2017, a warm-weather outlier at 73.9°F), we find that a single 5K checkpoint reduces root mean squared error by 876 seconds relative to the best pre-race model and narrows the 90% conformal interval from 77 to 55 minutes. A linear mixed-effects model with runner-specific random effects improves pre-race predictions for repeat runners, but this advantage vanishes once split data arrives. Upgrading from Ridge regression to gradient-boosted trees yields only a 56-second gain at 5K and no significant improvement at 40K. These results show that the information frontier, not model complexity, governs predictive accuracy.

I. Introduction

Information about a marathon runner arrives in stages. Age and sex are known at registration. Prior finish times become available from historical records. Once the race begins, cumulative split times arrive at each checkpoint. We study how predictive accuracy improves along this timeline and identify when each source stops providing meaningful gains. This complements the standard goal of minimizing prediction error by asking a different question: *where does predictive accuracy come from?*

We address three questions. First, how much does each information source (demographics, prior race history, cumulative checkpoint splits) contribute to predictive accuracy, and are these contributions statistically significant after accounting for within-runner dependence? Second, when does runner-specific personalization, modeled via linear mixed effects for repeat runners, become obsolete relative to live split data? Third, does switching from a linear model (Ridge regression) to a nonlinear model (gradient-boosted trees) improve accuracy beyond what the available information supports?

We make four contributions:

1) A nested five-model ablation with bias-corrected and accelerated (BCa) cluster-bootstrap confidence intervals that quantifies root mean squared error (RMSE) reductions from each information source.
2) Information decay curves that show how the marginal contribution of demographics, prior race history, and checkpoint splits diminishes as the race progresses.
3) Identification of a *crossover point*: the checkpoint, occurring as early as 5K, at which a Ridge regression model with split times produces a narrower conformal prediction interval than a personalized mixed-effects model.
4) A model-complexity benchmark showing that gradient-boosted trees do not improve over Ridge regression regression at later checkpoints. This indicates that the information frontier, not model complexity, governs accuracy.

II. Data Description

We use Boston Marathon results from 2000–2019, totaling 615,681 runner-race observations after preprocessing.[1] The preprocessing pipeline standardizes column formats across years, converts finish times to seconds, keeps only runners with recorded gender and age in $[14, 90]$, and removes duplicates.

[1]Raw data from 1897–2019 were obtained from https://github.com/adrian3/Boston-Marathon-Data-Project. We restrict to 2000–2019, the years for which checkpoint split times and weather data are available.

TABLE I: Dataset summary. All splits are strictly temporal.

Property	Value
Pre-race training rows	347,188
In-race training rows	52,779
Test rows	26,259
Test year	2017
Distinct runners	294,324
Repeat runners	65,590
Train mean max temp	55.0°F
Test max temp	73.9°F
Temperature gap	+18.9°F

Missing ages are imputed using Multivariate Imputation by Chained Equations (MICE), applied separately within each year with finish time and gender as auxiliary variables. Imputed values are clipped to $[14, 90]$ and rounded to integers. All downstream analyses exclude observations with imputed ages. The imputation step exists only to keep covariate matrices complete during the MICE procedure itself.

Race-day weather data (maximum temperature, mean humidity, mean wind speed) comes from the Meteostat API for Boston Logan Airport (station 72509) over a 10 AM–6 PM window. Logan Airport is about 10 miles from the course start in Hopkinton. We treat these measurements as a uniform proxy for on-course conditions.

Table I summarizes the temporal train/test split. All partitioning is strictly temporal with no random assignment.

a) Features.

Features fall into three groups, matching the ablation hierarchy in Section IV.

Demographics are basic runner characteristics: age, gender (binary female indicator), and race year.

Prior Boston history summarizes past performance. This includes the expanding-window mean of prior finish times, the number of prior appearances, and a log-transformed count $\log(1 + \text{prior_appearances})$. The log transform captures diminishing returns from additional races and remains defined for first-time runners ($k = 0$).

Checkpoint splits are cumulative race times at nine standard distances (5K, 10K, 15K, 20K, half-marathon, 25K, 30K, 35K, and 40K). Each additional checkpoint adds information about the runner's in-race performance.

b) Derived features (Light Gradient-Boosting Machine only).

The gradient-boosted tree models use additional engineered features not included in the Ridge regression models. These features involve interactions and ratios that tree ensembles can exploit but linear models cannot.

Pacing features describe how a runner's speed changes during the race: segment pace (time per segment between consecutive checkpoints), change in pace between consecutive segments, cohort-normalized pace z-scores (computed within gender × five-year age groups), and the variance of pace across completed segments.

Latent features serve as proxies for unobserved factors. Bib number is a noisy proxy for seeding tier and expected ability. It partially encodes qualifying times, so it is correlated with prior finish time but captures additional information about competitive standing. Heat exposure is the product of estimated race duration (projected from early pace) and maximum temperature.

III. Exploratory Data Analysis

A. Pacing Strategy

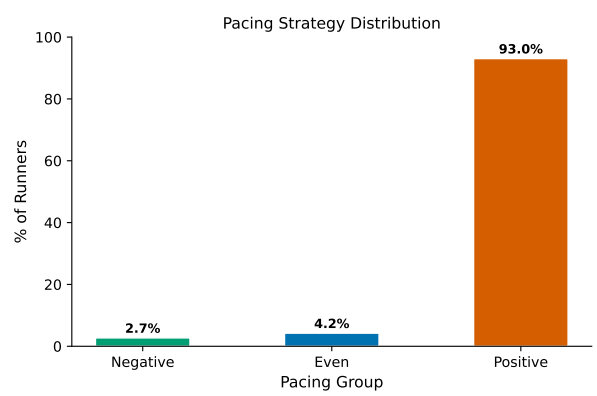


Fig. 1: Pacing strategy distribution across all runners in the in-race training and test sets. Positive splits dominate at 93.0%.

We classify pacing strategy by comparing each runner's second-half time to their first-half time. A ratio below 1 means the runner was faster in the second half ("negative split"). A ratio near 1 means even pacing. A ratio above 1 means the runner slowed in the second half ("positive split"). We use a 1% tolerance to define three groups: negative (< 0.99), even (0.99–1.01), and positive (> 1.01).

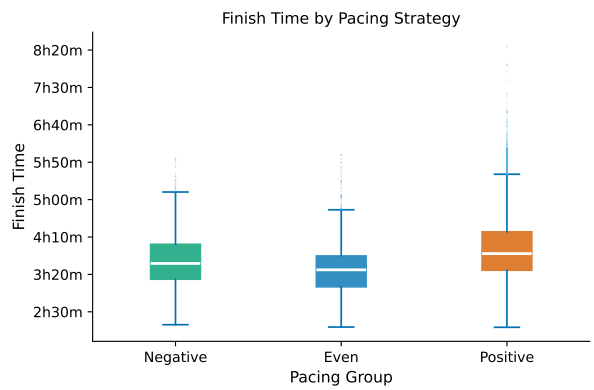


Fig. 2: Finish time by pacing strategy. Positive-split runners show greater variance and higher median finish times.

Figure 1 shows that pacing is highly asymmetric: 93.0% of runners positive-split, with only 4.2% even and 2.7% negative. Slowing in the second half is near-universal. This limits the range of pacing behaviors the models must handle.

Figure 2 shows the finish-time distributions for each group. Positive-split runners have higher median finish times and greater variance. Pacing strategy is therefore associated with both the level and the spread of finish times. However, the effect size is modest (eta-squared $\eta^2 = 0.019$, the proportion of variance explained, from a Kruskal–Wallis test, a nonparametric comparison of group medians): pacing alone explains only a small fraction of total variance.

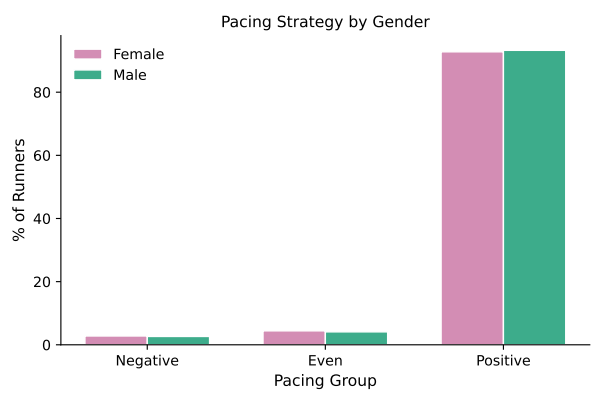


Fig. 3: Pacing strategy distribution by gender. Male and female runners show nearly identical pacing profiles.

Figure 3 shows that the pacing distribution is nearly identical for men and women. Both genders positive-split at approximately 93%, with similar proportions of even and negative splitters. This similarity supports using gender interaction terms within a single model rather than fitting separate models by gender.

B. Weather Impact

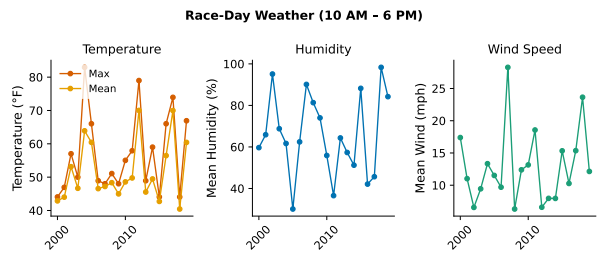


Fig. 4: Race-day weather at Boston Logan Airport (10 AM–6 PM) across 2000–2019. Temperature varies substantially across years; humidity and wind show no clear trend.

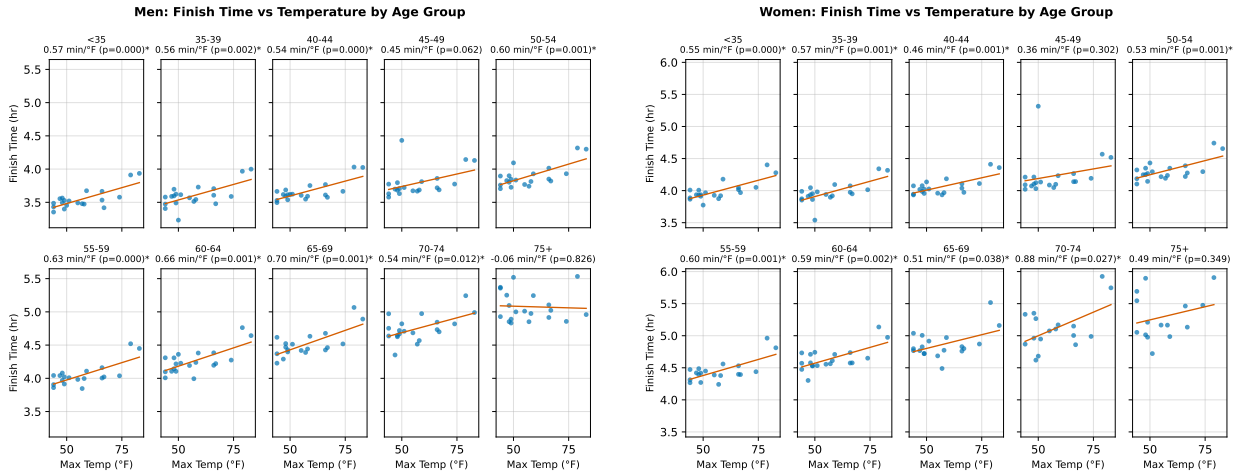
Figure 4 shows large year-to-year variation in race-day temperature. Humidity and wind show no consistent pattern. We therefore focus on temperature as the primary environmental variable.

We first test for a global association using Spearman rank correlation, which measures monotonic dependence without assuming linearity. Maximum temperature is strongly associated with mean finish time ($\rho = 0.661$, $p = 0.002$): hotter races are systematically slower. Mann–Whitney U tests (nonparametric two-sample comparisons) across age-gender groups find no significant effects for humidity or wind speed. This isolates temperature as the dominant weather factor.

We next measure the size of this effect. Per-age-group linear regressions (Figure 5) yield slopes of 0.45–0.70 minutes per degree Fahrenheit for men and 0.36–0.88 for women. These positive slopes are consistent across most age groups and represent a substantial effect.

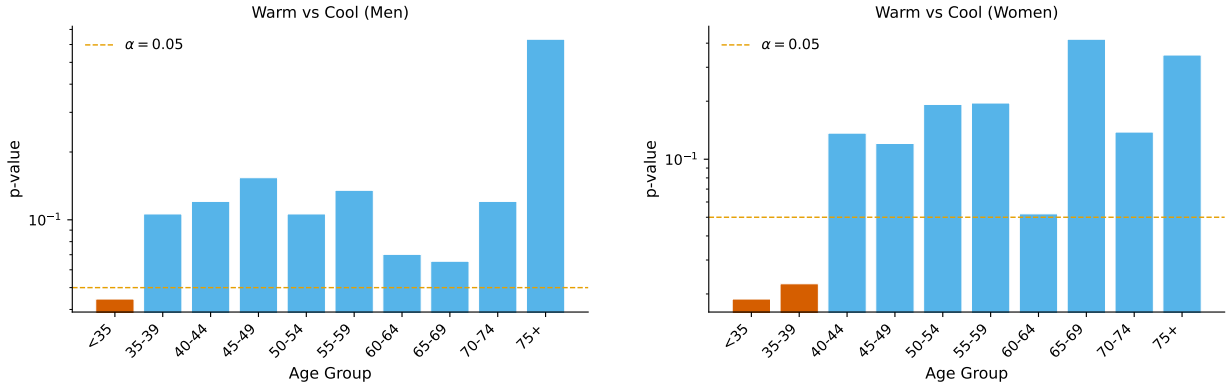
Finally, we check whether this effect holds across subpopulations. Figure 6 shows p-values for warm-versus-cool comparisons within each age group. Most groups fall below the $\alpha = 0.05$ threshold. Significance weakens in older age groups, likely due to smaller sample sizes rather than reduced heat sensitivity.

In summary, the temperature analysis shows three things: (i) a strong global association, (ii) a substantial effect size across age groups, and (iii) broad consistency with mild heterogeneity in older runners.



(a) Men (b) Women

Fig. 5: Per-age-group linear regression of mean finish time (hours) on maximum temperature (°F). Subplots report slopes (minutes per °F) and p-values; asterisks denote $p < 0.05$.



(a) Men (b) Women

Fig. 6: Per-age-group p-values comparing finish times between warm and cool race conditions. The horizontal line indicates $\alpha = 0.05$.

The 2017 test year had a maximum temperature of 73.9°F, which is 18.9°F above the training-period mean. This creates a natural distributional shift that makes 2017 a demanding test of model robustness.

C. Implications for Modeling

Both pacing and temperature analyses reveal population-level structure that in-race split data ultimately absorbs. Pacing behavior is directly encoded in the cumulative splits. Temperature effects show up as slower checkpoint times on warm days. This previews the central finding of the ablation in Section VI: once split data is available, pre-race features provide diminishing returns.

IV. Methodology

We fit five models in sequence, each using more information than the last. Each model adds one information source to the previous model. Comparing RMSE between adjacent models isolates the marginal contribution of that source. After defining the model sequence, we describe two shared inference procedures: conformal prediction intervals (Section IV-G) and BCa cluster-bootstrap confidence intervals (Section IV-H).

A. Sample Mean

The baseline predicts every runner's finish time with the training-set mean:

$$\hat{y}_i = \bar{y}_{\text{train}} = \frac{1}{n}\sum_{j=1}^{n} y_j. \tag{1}$$

This is the best constant predictor under squared error loss. It uses no information about individual runners and serves as the ceiling against which all subsequent models are compared.

B. Ordinary Least Squares (OLS) with Demographic Covariates

The first informative model uses demographic information available at registration. It does not use any in-race data. We regress log-transformed finish time on centered covariates with nonlinear age effects and gender interactions:

$$\log(s_i) = \beta_0 + \beta_1 a_i + \beta_2 a_i^2 + \beta_3 f_i + \beta_4 t_i + \beta_5 (a_i \cdot f_i) + \beta_6 (a_i^2 \cdot f_i) + \varepsilon_i, \tag{2}$$

where s_i is finish time in seconds, $a_i = \text{age}_i - \bar{a}_{\text{train}}$ is mean-centered age, $f_i \in \{0, 1\}$ is a female indicator, and $t_i = \text{year}_i - 2008$ is a centered calendar-year trend.

a) Why model on the log scale.

Marathon finish times are right-skewed: most runners finish in a moderate range, but a smaller number of much slower runners stretch the distribution. This skew violates the OLS assumption that errors are symmetric. The log transform compresses large values, producing a more symmetric distribution. It also stabilizes the variance. Without the transform, slower runners show much larger residual spread than faster runners. This violates the constant-variance assumption. The log scale fixes both problems and improves the reliability of the estimates.

b) Why center the covariates.

Centering subtracts the training-set mean from each continuous predictor. This has two benefits. First, the intercept β_0 becomes interpretable: it is the predicted log finish time for a runner of average age in the reference year (2008). Without centering, β_0 would correspond to a hypothetical zero-year-old. Second, centering reduces collinearity between a_i and a_i^2. Without centering, these two terms are correlated because squaring positive numbers preserves the ranking. This correlation inflates the variance of their coefficient estimates. Centering decorrelates them and produces more stable estimates.

c) Why include quadratic and interaction terms.

The quadratic term $\beta_2 a_i^2$ lets the model capture a U-shaped relationship between age and performance. Finish times decrease through peak competitive years and then increase with age-related decline. Without this term, the model could only represent a straight-line relationship and would miss the curvature. The gender interaction terms $\beta_5(a_i \cdot f_i)$ and $\beta_6(a_i^2 \cdot f_i)$ let the peak age and rate of decline differ between men and women. Without these terms, the model would assume identical aging patterns for both genders.

d) Why use HC3 (heteroskedasticity-consistent, variant 3) robust standard errors.

Standard Ordinary least squares (OLS) assumes that every observation has the same error variance. In our data, this does not hold: slower runners have more variable finish times than faster runners. When this assumption fails, classical standard errors can be wrong.

HC3 fixes this by estimating a separate variance for each observation based on its squared residual. It also corrects for high-leverage observations, where residuals appear artificially small because the model has partially fitted to that point. We chose HC3 over simpler alternatives (HC0, HC1, HC2) because it performs better in finite samples.

e) Why use bias-correction retransformation.

The model is fitted on the log scale, so predictions must be transformed back to seconds. Naive exponentiation systematically underpredicts finish times because of Jensen's inequality: $\mathbb{E}[\exp(\varepsilon_i)] > 1$ even when $\mathbb{E}[\varepsilon_i] = 0$. The bias-correction factor [10, 11] corrects this:

$$\hat{y}_i = \exp(\mathbf{x}_i^\top \hat{\boldsymbol{\beta}}) \cdot D, \quad D = \frac{1}{n}\sum_{j=1}^{n} \exp(\hat{\varepsilon}_j), \tag{3}$$

where $\hat{\varepsilon}_j = \log(s_j) - \mathbf{x}_j^\top \hat{\boldsymbol{\beta}}$ is the j-th residual on the log scale. The bias-correction factor D estimates $\mathbb{E}[\exp(\varepsilon_i)]$ nonparametrically, making no assumption about the error distribution. This is preferable to

the parametric correction $\exp(\hat{\sigma}^2/2)$, which requires normally distributed errors [10].

C. OLS with History Covariates

We add prior Boston Marathon performance to the demographic model. This represents the richest pre-race information available:

$$\log(s_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \beta_7 \log(1 + k_i) + \beta_8 \bar{s}_i^{\text{prior}} + \varepsilon_i, \qquad (4)$$

where $\mathbf{x}_i^\top \boldsymbol{\beta}$ contains the demographic terms from (2), k_i is the number of prior Boston finishes, and $\bar{s}_i^{\text{prior}}$ is the expanding-window mean of previous finish times.

a) Why use $\log(1 + k_i)$ instead of k_i.

The log transform captures diminishing returns. The difference between one and two prior races is more informative than the difference between ten and eleven. The +1 ensures the term is defined for first-time runners ($k_i = 0$), where $\log(1 + 0) = 0$.

b) Why use an expanding window.

The expanding-window mean uses only races that occurred before the current one. This prevents data leakage. If we included future performances in the average, the model would have access to information that would not be available at prediction time.

c) Handling first-time runners.

For first-time runners, $k_i = 0$ and $\bar{s}_i^{\text{prior}}$ is undefined. In the ablation evaluation, these runners receive predictions from the demographics-only model (2). This ensures all models in the hierarchy are evaluated on the same test population.

Estimation, bias-correction retransformation (3), and leave-one-out (LOO) residuals (**??**) follow the same procedures as Section IV-B.

d) Limitation.

This model represents history through fixed summary features and treats repeated observations from the same runner as independent. It cannot capture unobserved runner-specific effects or differences in how runners age over time. This motivates the mixed-effects extension below.

D. Linear Mixed-Effects Model

The OLS models treat each observation as independent. However, our data contains repeated measurements from the same runner across multiple years. A linear mixed-effects (LME) model [2, 3] accounts for this by introducing runner-specific random effects.

We fit the model on repeat runners (two or more appearances in 2000–2016):

$$\log(s_{ij}) = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + u_{0i} + u_{1i} \cdot a_{ij} + \varepsilon_{ij}, \qquad (5)$$

where i indexes runners, j indexes race appearances, $\mathbf{x}_{ij}^\top \boldsymbol{\beta}$ contains the fixed effects from (2), u_{0i} is a random intercept capturing whether runner i is persistently faster or slower than average, and u_{1i} is a random slope on centered age allowing the rate of decline to vary across runners. The random effects follow:

$$\begin{bmatrix} u_{0i} \\ u_{1i} \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0},\ \boldsymbol{\Sigma} = \begin{bmatrix} \sigma^2_{u_0} & \sigma_{u_0 u_1} \\ \sigma_{u_0 u_1} & \sigma^2_{u_1} \end{bmatrix}\right), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2),$$

with $\varepsilon_{ij} \perp (u_{0i}, u_{1i})$(errors are independent of the random effects). The off-diagonal element $\sigma_{u_0 u_1}$ captures the relationship between baseline ability and aging rate. We estimate parameters via restricted maximum likelihood (REML; Bates et al. 2), which produces less biased variance component estimates than standard maximum likelihood.

a) Variance decomposition.

We report marginal and conditional R^2 [16]:

$$R^2_{\text{marg}} = \frac{\text{Var}(\mathbf{X}\hat{\boldsymbol{\beta}})}{\text{Var}_{\text{total}}}, \quad R^2_{\text{cond}} = \frac{\text{Var}(\mathbf{X}\hat{\boldsymbol{\beta}}) + \text{Var}(\mathbf{Z}\hat{\mathbf{u}})}{\text{Var}_{\text{total}}}, \qquad (6)$$

where $\text{Var}_{\text{total}}$ sums fixed-effect, random-effect, and residual variance components. The marginal R^2 measures variance explained by demographics alone. The conditional R^2 adds runner-specific effects. In our data, $R^2_{\text{marg}} = 0.198$ versus $R^2_{\text{cond}} = 0.768$ (Table VI). Most explainable variation is runner-specific rather than demographic.

b) Shrinkage.

Most runners appear only 2–3 times (median 2, IQR [2, 3]). With so few observations per runner, the model pulls individual estimates toward the population mean [3]. We measure this as $\text{Var}(\hat{u}_{0i})/\text{Var}(\bar{r}_i)$, where $\bar{r}_i$ is the raw per-runner residual mean. The observed ratio is 0.443 (Table VI): the model pulls estimates nearly halfway back to the population mean. This limits the

practical benefit of personalization even before in-race data arrives.

c) Prediction.

Marginal predictions use only fixed effects and apply to never-seen runners. Conditional predictions add the best linear unbiased predictors (BLUPs) $\hat{u}_{0i}$ and $\hat{u}_{1i}$ [2] and apply to known repeat runners. Because BLUPs are shrinkage estimators, conditional predictions for runners with few appearances stay close to the marginal prediction. Both are retransformed using the bias-correction factor (3).

d) Conformal calibration.

The model is trained through 2015. The calibration threshold $\hat{q}$ (Section IV-G) is calibrated on 2016 prediction errors for known runners, then evaluated on the 2017 test set.

E. Ridge Regression at Checkpoints

Once the race begins, cumulative split times give direct information about a runner's performance. These features are strongly collinear: the cumulative time at 10K includes the time at 5K. This makes Ordinary least squares (OLS) unstable, so we use Ridge regression [4], which adds an ℓ_2 (sum-of-squares) penalty to stabilize coefficient estimates.

At each checkpoint $c \in \{5\text{K}, 10\text{K}, 15\text{K}, 20\text{K}, \text{Half}, 25\text{K}, 30\text{K}, 35\text{K}, 40\text{K}\}$, we solve:

$$\hat{\boldsymbol{\beta}}_c = \arg\min_{\boldsymbol{\beta}} \left\{ \|\mathbf{y} - \mathbf{X}_c \boldsymbol{\beta}\|^2 + \lambda_c \|\boldsymbol{\beta}\|^2 \right\}, \tag{7}$$

where $\mathbf{y}$ is the vector of finish times (in seconds, on the original scale) and $\mathbf{X}_c$ contains cumulative splits through checkpoint c plus age and a female indicator. The first term measures prediction error. The second penalizes large coefficients, preventing the model from assigning extreme weights to correlated features. The parameter $\lambda_c \geq 0$ controls the trade-off [4]: when $\lambda_c = 0$, Ridge regression reduces to OLS; as λ_c increases, coefficients shrink toward zero, trading bias for stability.

Unlike the OLS models, Ridge regression operates on the original scale rather than the log scale. The Ridge regression penalty directly addresses collinearity, making log-scale variance stabilization unnecessary.

We select λ_c via generalized cross-validation (GCV) [4, 12], which approximates LOO error without refitting the model n times. LOO residuals are available in closed form through the Ridge regression projection matrix (the matrix that maps observed responses to fitted values):

$$e_{(-i)} = \frac{y_i - \hat{y}_i}{1 - h_{ii}}, \tag{8}$$

and are used to compute the calibration threshold $\hat{q}_c$ as described in Section IV-G.

F. Light Gradient-Boosting Machine at Checkpoints

Ridge regression assumes that the relationship between features and finish time is additive and linear. This may miss important structure. For example, the effect of a fast early pace on finish time might depend on age (an interaction). Or the mapping from pace to finish time might be curved (a nonlinearity).

a) Why use gradient-boosted trees.

We fit gradient-boosted decision trees (Light Gradient-Boosting Machine; Ke et al. 6) at each checkpoint as a nonlinear benchmark. Gradient boosting builds a sequence of small decision trees. Each tree is trained to correct the errors of the trees before it. Each tree can capture interactions by splitting on different features at different nodes. It can also capture nonlinearities by using piecewise-constant approximations.

This model serves as a complexity ceiling. If it does not substantially outperform Ridge regression, then the mapping from splits to finish time is approximately linear. In that case, accuracy is limited by the information available, not by model capacity.

We use fixed hyperparameters (500 trees, learning rate 0.01, max depth 4, leaf-wise growth) without tuning.

b) Why not tune hyperparameters.

This is deliberate. Tuning would mix two effects: the benefit of a more flexible model and the benefit of a better configuration search. By fixing the hyperparameters, we isolate the first effect. Any improvement over Ridge regression is then due to the model family itself, not to tuning.

c) Why extend the feature set for Light Gradient-Boosting Machine only.

Light Gradient-Boosting Machine receives the same checkpoint splits and demographics as Ridge regression, plus additional engineered features:

- *Segment paces*: time between consecutive checkpoints, measuring speed in each portion of the race;
- *Pace changes*: differences in pace between consecutive segments, capturing acceleration or deceleration;
- *Cohort-normalized pace z-scores*: each runner's pace expressed as standard deviations from the mean of runners of similar age and gender;
- *Pace variance*: variance of segment paces across completed segments, measuring pacing consistency;
- *Bib number*: a proxy for seeding tier, since Boston Marathon bibs are assigned based on qualifying times;
- *Heat exposure*: the product of projected race duration and maximum temperature, capturing the interaction between race length and thermal stress.

These features involve ratios, interactions, and within-cohort comparisons that a linear model cannot use without manual feature engineering. We withhold them from Ridge regression to keep the comparison clean. If we gave both models the same extended features, we could not tell whether any improvement came from the model class or from the extra features. Giving them only to Light Gradient-Boosting Machine gives the nonlinear model every advantage. If it still fails to improve, the conclusion that model complexity does not matter is stronger.

d) Why use split-conformal instead of LOO.

The closed-form LOO shortcut (8) relies on the projection matrix [4], which exists only for linear models. For tree ensembles, no such formula is available. Refitting the model n times is too expensive.

We instead use split-conformal calibration [7]. We hold out 20% of the training data as a calibration set (with a fixed random seed for reproducibility). The model is trained on the remaining 80%. The calibration threshold $\hat{q}_c$ is computed from calibration-set residuals as in Section IV-G. Split-conformal uses less training data than LOO, but it works for any model class.

G. Conformal Prediction

Each model in the hierarchy produces point predictions. To measure prediction uncertainty, we build prediction intervals using conformal inference [1, 7, 14].

a) Why use conformal prediction.

Traditional prediction intervals require distributional assumptions. For example, they may assume normally distributed errors. If these assumptions are wrong, the intervals may be too narrow or too wide. Conformal prediction avoids this problem. It guarantees that the interval contains the true value with at least the specified probability. The only assumption is that calibration and test observations are exchangeable [1, 14]. Roughly, this means they come from the same distribution.

b) Procedure.

Given n out-of-sample absolute residuals $|y_i - \hat{y}_{(-i)}|$ (from LOO for linear models, or from a held-out calibration set for Light Gradient-Boosting Machine), the calibration threshold is:

$$\hat{q} = \text{quantile}\left(\left|y_i - \hat{y}_{(-i)}\right|, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right), \quad \alpha = 0.10. \tag{9}$$

The quantile level $\lceil (n+1)(1-\alpha) \rceil / n$ is slightly above $1-\alpha$. This is a finite-sample correction [14]. With a finite number of calibration points, using the exact 0.90 quantile can produce undercoverage. The adjusted level guarantees at least $1-\alpha = 90\%$ coverage even with a limited calibration set.

The prediction interval for a new observation is:

$$[\hat{y}_i - \hat{q},\ \hat{y}_i + \hat{q}].$$

This interval is symmetric around the point prediction and has the same width $2\hat{q}$ for every runner [7].

c) Why report interval width.

We report $2\hat{q}$ (in minutes) alongside RMSE as a primary metric. RMSE measures point-prediction accuracy. The interval width $2\hat{q}$ measures calibrated precision: a narrower interval at the same coverage level means the model is extracting more information from the available features. We use $2\hat{q}$ to identify the *personalization*

crossover point: the earliest checkpoint at which the Ridge regression model's interval is narrower than the best pre-race model's. This marks the point where in-race data surpasses all pre-race information in predictive value.

H. BCa Cluster-Bootstrap Confidence Intervals

We test whether observed RMSE differences between adjacent models are statistically significant using bias-corrected and accelerated (BCa) bootstrap confidence intervals [5]. We resample at the runner level (cluster bootstrap; MacKinnon et al. 8, 9;Wicklin 15) to preserve within-runner dependence.

We draw $B = 1{,}000$ bootstrap samples by resampling runners with replacement. For each sample b, we recompute the RMSE difference $\hat{\theta}_b^*$. The BCa method improves on the standard percentile bootstrap by correcting for bias and skewness.

The acceleration parameter corrects for skewness using jackknife empirical likelihood for the skewness and kurtosis of influence values (Wicklin 15), $d_i = \bar{d} - d_{(i)}$, where $d_{(i)}$ is the RMSE difference with runner i removed:

$$a = \frac{\sum_{i=1}^{m} d_i^3}{6\left(\sum_{i=1}^{m} d_i^2\right)^{3/2}}. \tag{10}$$

The bias correction [5] measures whether the bootstrap distribution is centered at the observed estimate:

$$z_0 = \Phi^{-1}\left(\mathrm{clip}\left(\frac{\#\{\hat{\theta}_b^* < \hat{\theta}\}}{B}, \frac{1}{B+1}, \frac{B}{B+1}\right)\right). \tag{11}$$

Both corrections are combined into adjusted percentile levels:

$$p_{\mathrm{lo/hi}} = \Phi\left(z_0 + \frac{z_0 + z_{\alpha/2}}{1 - a(z_0 + z_{\alpha/2})}\right), \tag{12}$$

where $z_{\alpha/2} = \Phi^{-1}(\alpha/2)$. When $z_0 = 0$ and $a = 0$, this reduces to the standard percentile method. The final confidence interval is $[\hat{\theta}^*_{(p_{\mathrm{lo}})}, \hat{\theta}^*_{(p_{\mathrm{hi}})}]$. A 95% BCa interval that excludes zero provides evidence that the RMSE difference is statistically significant.

a) Computational efficiency.

We pre-aggregate per-runner squared errors before bootstrapping [9] [15]. Each bootstrap sample then only needs to resample these runner-level aggregates and compute a weighted sum. This avoids re-evaluating model predictions and reduces 1,000 bootstrap iterations to a few seconds.

V. Experimental Setup

This section defines how the models are trained, evaluated, and compared. The setup follows the information timeline from Section IV. Each model is trained and evaluated using only the information available at its stage.

a) Temporal train/test split.

All data is split by time, with no random assignment. Pre-race models (demographics, history, mixed effects) are trained on 2000–2016 (347,188 observations). In-race models (Ridge regression and Light Gradient-Boosting Machine) are trained on 2015–2016 (52,779 observations). These are the years for which checkpoint split data is available. All models are evaluated on the 2017 race (26,214 runners with complete splits through 40K).

The pre-race models have about seven times more training data than the in-race models. This is not a design choice. Split times were not systematically recorded before 2015. Because the in-race models achieve much lower RMSE despite having less training data, this asymmetry works against our central finding. It makes the information-hierarchy results conservative.

b) Evaluation metrics.

We evaluate all models on the same 2017 test set using two metrics. RMSE (in seconds) measures point-prediction accuracy. It is more sensitive to large errors than to small ones. Conformal interval width $2\hat{q}$ (in minutes) measures calibrated precision: how tightly the model can bound a runner's finish time while maintaining 90% coverage. Reporting both ensures that improvements in point prediction come with genuine reductions in uncertainty.

c) Nested ablation.

The primary evaluation compares adjacent models in the information hierarchy. For each consecutive pair, we compute the RMSE reduction (ΔRMSE) with 95% BCa cluster-bootstrap confidence intervals ($B = 1{,}000$; MacKinnon et al. 8). All pairwise differences are computed within a single bootstrap run. This preserves the dependence structure across models and keeps the confidence intervals internally consistent.

d) Information decay analysis.

To measure how the value of pre-race features changes as in-race data accumulates, we run a controlled ablation at each Ridge regression checkpoint. The procedure separates the contribution of added features from the effect of regularization:

1) Fit a base Ridge regression model using only checkpoint splits. Select λ_c^* via Generalized Cross-Validation (GCV).
2) Fit two augmented Ridge regression models: one adding demographic features, one adding prior history features. Keep λ_c^* fixed at the value from step 1 for both.
3) Report the RMSE reduction from each augmented model as the marginal contribution of the added features at checkpoint c.

Fixing λ_c^* is important. It ensures that any change in RMSE is due to the added features, not to a change in regularization strength. If λ were re-optimized, the augmented model might achieve lower RMSE partly because the optimizer found a better penalty value. This would mix the feature contribution with the regularization effect.

e) Model-family comparison.

We compare Ridge regression and Light Gradient-Boosting Machine at each checkpoint to test whether a nonlinear model improves accuracy beyond what the available information supports. The two models use different conformal calibration procedures (LOO for Ridge regression, split-conformal for Light Gradient-Boosting Machine), so their bootstrap samples are not naturally paired. We therefore use conservative confidence intervals based on independent bootstrap bounds. These produce wider intervals and avoid overstating the significance of any difference.

f) Reproducibility.

All random number generation uses independent, seeded streams for each stage of the analysis. Ridge regression uses deterministic Generalized Cross-Validation (GCV). Light Gradient-Boosting Machine runs in deterministic mode. Complete code is available at https://github.com/indrajeetadityaroy9/Boston_Marathon_Project.

VI. Results

Results follow the information timeline from Section IV. Each subsection shows how prediction improves as more information becomes available and whether the gains are statistically significant. All models are evaluated on the same 2017 test set ($n = 26{,}214$) unless otherwise noted.

A. Nested Information Hierarchy

Table II and Figure 7 present the full ablation. Each row adds one information source to the previous model. The table reads as a cumulative sequence. The columns are:

- **RMSE** (seconds) is the typical prediction error. It is the square root of the mean squared difference between predicted and observed finish times. Lower values mean more accurate predictions.
- **ΔRMSE** is the RMSE reduction from adding one information source. It measures the improvement over the previous row. The 95% CI is a bias-corrected and accelerated (BCa) bootstrap confidence interval that accounts for within-runner dependence (Section IV-H).
- **Coverage** is the fraction of true finish times that fall inside the 90% conformal prediction interval. Values near 90% mean the intervals are well calibrated.
- $2\hat{q}$ (minutes) is the full width of the conformal prediction interval. Smaller values mean more precise predictions. The model can bound a runner's finish time within a narrower range.

Stage	Description	N	RMSE	ΔRMSE	95% CI	Coverage (%)	$2\hat{q}$ (min)
Sample Mean	Constant prediction	26214	2538.0	–	–	–	–
OLS Demographics	+ demographics (HC3)	26214	2317.3	220.7	[208.2, 233.4]	89.2	118.4
OLS Demogr.+History	+ prior history (HC3)	26214	2126.8	190.5	[177.5, 203.0]	89.7	82.5
Ridge 5K	+ 5K splits (GCV)	26214	1250.8	876.0	[849.3, 902.0]	85.8	54.6
Ridge 40K	+ 40K splits (GCV)	26214	123.0	1127.9	[1093.2, 1157.0]	86.4	2.8

TABLE II: Nested information hierarchy. Each row adds one information source. ΔRMSE is the improvement over the previous row (BCa 95% CI). Coverage and $2\hat{q}$ are from conformal calibration at $\alpha = 0.10$.

The sample mean predicts every runner with a single constant. It has an RMSE of 2538.0 seconds (about 42 minutes). Adding demographics reduces RMSE by 220.7 seconds. Adding prior race history gives a further 190.5-second improvement. Together, these two pre-race stages reduce RMSE by 411 seconds.

The largest single improvement comes from the first 5K split. The Ridge regression model at 5K reduces RMSE by 876.0 seconds [849.3, 902.0]. This one checkpoint more than doubles the combined gain from all pre-race

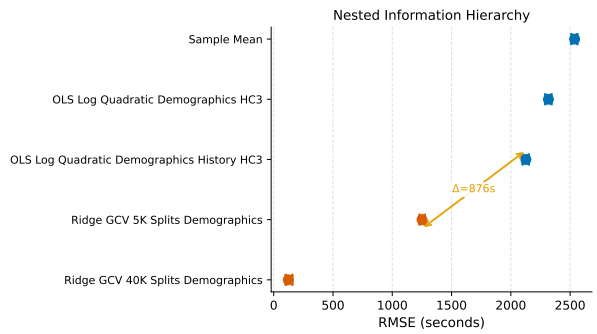


Fig. 7: RMSE across the information hierarchy. Error bars show BCa 95% confidence intervals. The arrow marks the 876-second improvement from the best pre-race model to the 5K ridge model.

information (876 vs. 411 seconds). The step change is visible in Figure 7.

From 5K to 40K, RMSE drops further to 123.0 seconds (about 2 minutes). The interval width $2\hat{q}$ shrinks from 54.6 to 2.8 minutes. All five models show a monotonic decrease in RMSE, and all pairwise ΔRMSE confidence intervals exclude zero. This confirms that each information source provides a statistically significant, non-redundant contribution.

a) Undercoverage at in-race checkpoints.

Pre-race models achieve coverage close to the 90% target (89.2% and 89.7%). The in-race Ridge regression models show undercoverage: 85.8% at 5K and 86.4% at 40K. This means roughly 1 in 7 runners falls outside the prediction interval. The nominal target is 1 in 10. Pre-race models maintain near-nominal coverage because their wider intervals buffer against the distributional shift. The sharper in-race intervals are more exposed.

The most likely cause of the undercoverage is the distributional shift between training and test conditions. The 2017 test race was 18.9°F warmer than the training average. This violates the exchangeability assumption underlying the conformal coverage guarantee [13, 14]. Runners in extreme heat produce larger residuals than the calibration set expects. This widens the gap between nominal and observed coverage. Temperature-adaptive or conditional conformal methods could help close this gap.

B. Personalization Crossover

We next ask two questions. Does runner-specific personalization help beyond population-level patterns? And at what point does in-race data surpass personalized pre-race prediction?

Table III compares two pre-race models on repeat runners (n = 4,638). The first is an OLS model using demographics and prior history with HC3 robust standard errors. The second is a linear mixed-effects (LME) model that adds runner-specific random effects: a random intercept and a random age slope. The LME model pools each runner's predictions partly toward their own history. RMSE is reported with BCa 95% confidence intervals. ΔRMSE is the improvement of the LME model over OLS. The lower baseline RMSE on repeat runners (1469.6 vs. 2126.8 on the full test set in Table II) reflects the fact that runners with prior history are inherently more predictable.

Model	RMSE (s) [95% CI]
OLS Demogr.+History (HC3)	1469.6 [1420.5, 1526.8]
LME Conditional (RI+Age)	1400.0 [1347.9, 1457.4]
ΔRMSE	+69.6 [+57.7, +83.7]

TABLE III: Personalization comparison on repeat runners (n = 4,638). ΔRMSE is the RMSE reduction from the LME model relative to OLS (BCa 95% CI).

The mixed-effects model reduces RMSE by 69.6 seconds [57.7, 83.7]. The confidence interval excludes zero, so the improvement is statistically significant. Runner-specific effects help pre-race prediction, but the gain is small compared to the 876-second improvement from the first 5K split.

Table IV places both pre-race models alongside the 5K Ridge regression model. All three are evaluated on the same repeat-runner subset. The crossover point is the first checkpoint where in-race data produces a narrower conformal interval than the best personalized pre-race model.

Model	RMSE	Cov. (%)	$2\hat{q}$ (min)
OLS Demogr.+History (HC3)	1469.6	91.7	82.5
LME RI+Age Slope	1400.0	91.4	77.2
Ridge 5K+Demogr.	1250.8	85.8	54.6

TABLE IV: Conformal crossover on repeat runners. The 5K ridge model produces the narrowest prediction interval despite using no runner-specific history.

The key comparison is the interval width $2\hat{q}$. The OLS model has width 82.5 minutes. The mixed-effects model improves this to 77.2 minutes. The 5K Ridge

regression model reduces it to 54.6 minutes. That is 22.6 minutes narrower than the best personalized pre-race model. This crossover happens at the earliest available checkpoint. Even a few kilometers of in-race data provide stronger constraints on prediction than years of personal race history. This gives prediction systems a clear switching rule: adopt the in-race model as soon as split data is available.

a) Robustness of the information takeover.

Table V tests this result with three 5K Ridge regression variants, compared against the best pre-race model on the full test set. The column Δ vs Pre-Race is the RMSE difference (in seconds) relative to the best pre-race OLS model. The three variants differ in how they use prior information. The base model includes only demographics. The second adds runner-specific random effects for repeat runners. The third is restricted to runners with no prior Boston Marathon history.

Model	RMSE (s)	Δ vs Pre-Race (s)
Best pre-race (OLS, HC3)	2126.8	—
Ridge 5K+Demographics	1250.8	−876.0
Ridge 5K + Runner RE	1180.6	−946.2
Ridge 5K (no prior history)	1270.8	−856.0

TABLE V: Information takeover at 5K. All three in-race variants substantially outperform the best pre-race model.

The base 5K Ridge regression model reduces RMSE by 876 seconds. Adding runner random effects gives a further 70-second improvement (1250.8 vs. 1180.6). This shows that personalization still has modest value when combined with in-race data. But even runners with no prior Boston history reach RMSE of 1270.8 seconds. That is only 20 seconds worse than the full model and 856 seconds better than the best pre-race prediction. In-race data alone is enough to beat all pre-race models, whether or not the runner has been seen before.

b) Variance decomposition.

Table VI explains why personalization has limited impact.

Fixed effects explain only 19.8% of the variance in log finish times. Adding random effects raises this to 76.8%. Most of the predictable variation is runner-specific rather than demographic. It comes from persistent individual differences in ability. These are exactly the differences that cumulative split times reveal once the race begins. This is why in-race data so quickly makes pre-race personalization redundant.

Metric	Value
Fixed-effects R^2	0.198
Fixed + Random R^2	0.768
Shrinkage ratio	0.443
Appearances per runner (median [IQR])	2 [2, 3]
Marginal RMSE (known runners)	2156.8
Conditional RMSE (known runners)	1535.1
Marginal RMSE (never-seen runners)	2264.0
$2\hat{q}_{\mathrm{LME}}$ (min)	77.2

TABLE VI: Mixed-effects variance decomposition. Fixed-effects R^2 is the variance explained by demographics alone. Fixed + Random R^2 adds runner-specific effects. The shrinkage ratio measures how strongly individual estimates are pulled toward the population mean. Values near 0 mean strong pooling. Values near 1 mean individual data dominates.

The shrinkage ratio of 0.443 shows a second limitation. Most runners appear only 2–3 times. The model cannot reliably tell individual ability apart from race-to-race noise. It pulls individual estimates nearly halfway back toward the population mean. This limits how precise personalized predictions can be.

The conditional RMSE for known runners (1535.1 seconds) confirms this. Even with the best possible runner-specific estimates, the pre-race prediction is still far above the 5K Ridge regression RMSE of 1250.8 seconds. Personalization narrows the gap between known and unknown runners (1535.1 vs. 2264.0). But it cannot close the gap between pre-race and in-race prediction.

C. Information Decay

We next measure how the value of pre-race features changes as in-race data accumulates. At each checkpoint, we fit three matched-λ Ridge regression models: one with split times only, one adding demographics, and one adding prior history. All three use the same regularization parameter λ_c^*, selected via Generalized Cross-Validation (GCV) on the splits-only model. This ensures that any RMSE reduction comes from the added features, not from a change in regularization (Section V).

Table VII reports ΔRMSE: the RMSE reduction from adding each feature group on top of splits alone. Large values mean the feature group still carries useful information beyond the observed splits. Values near zero mean the splits have absorbed the feature's predictive content.

Checkpoint	Demographics ΔRMSE [95% CI]	Prior History ΔRMSE [95% CI]
5K	8.5 [7.0, 9.9]	24.9 [19.2, 33.1]
10K	11.4 [9.8, 13.0]	13.3 [10.1, 18.5]
15K	15.2 [13.4, 17.1]	6.9 [4.9, 9.5]
20K	13.1 [11.6, 14.5]	3.2 [1.7, 4.7]
HALF	10.9 [9.6, 12.2]	2.7 [1.3, 4.4]
25K	9.7 [8.6, 10.8]	1.7 [0.8, 2.8]
30K	5.7 [4.9, 6.6]	0.7 [0.3, 1.3]
35K	2.5 [2.0, 3.1]	0.2 [0.1, 0.4]
40K	0.6 [0.3, 0.9]	0.0 [0.0, 0.1]

TABLE VII: Information decay across checkpoints. ΔRMSE is the marginal contribution of each feature group under matched-λ Ridge regression (BCa 95% CI).

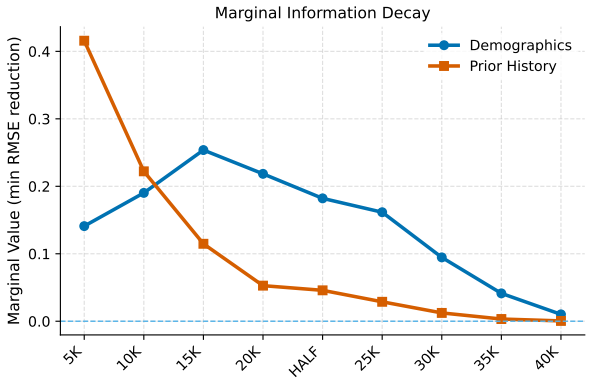


Fig. 8: Marginal RMSE reduction from demographics and prior history at each checkpoint. Prior history starts higher but decays faster. The two curves cross between 10K and 15K.

The two feature groups decay in different ways (Figure 8).

Prior history contributes 24.9 seconds at 5K but decays rapidly. Its contribution is halved by 10K (13.3 seconds). It reaches 2.7 seconds at the half-marathon and is effectively zero at 40K (0.0 seconds, [0.0, 0.1]). This makes sense: historical finish times are a proxy for persistent ability. Early split times reveal ability directly. Once the model sees a runner's current pace, the historical average becomes redundant.

Demographics follow a different pattern. Their contribution rises from 8.5 seconds at 5K to a peak of 15.2 seconds at 15K, then declines to 0.6 seconds at 40K. This inverted-U shape reflects a timing mismatch. Age and sex predict second-half fatigue patterns. Older runners and women tend to slow differently in the later miles. But these effects are not yet visible from the first few kilometers of pace data. In early checkpoints, the model has little information about how the runner will fatigue. Demographic features temporarily fill this gap by predicting future deceleration. Beyond 15K, the observed splits capture the runner's fatigue trajectory directly. The demographic contribution then diminishes.

The two curves cross between 10K and 15K (Figure 8). Before this point, history is more valuable than demographics. After it, demographics are more valuable. This crossover reflects the different types of information each source provides. History is a direct proxy for ability and is quickly replaced by observed ability. Demographics encode structural relationships between runner characteristics and fatigue. These take longer for the accumulating splits to absorb.

By 40K, both sources contribute less than 1 second of RMSE reduction. The full split vector absorbs nearly all predictive content from pre-race features.

Figure 9 shows the same pattern from the uncertainty side. The conformal interval width shrinks from 54.6 minutes at 5K to 2.8 minutes at 40K. This is a twenty-fold narrowing. Both accuracy and calibrated precision improve as in-race data grows.

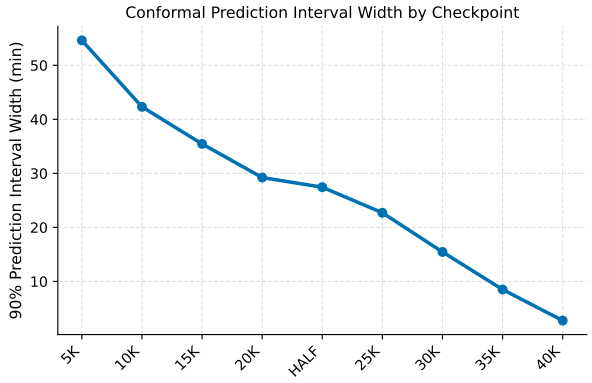


Fig. 9: Conformal prediction interval width ($2\hat{q}$) across checkpoints. The interval narrows by a factor of 20 from 5K to 40K.

D. Model Complexity as Complexity Benchmark

Finally, we test whether a more complex model improves accuracy beyond what the available information supports. Table VIII and Figure 10 compare Ridge regression [4] (a linear model with ℓ_2 regularization) and Light Gradient-Boosting Machine [6] (a gradient-boosted decision tree ensemble that can capture non-linear relationships and feature interactions).

Both models receive the same core features: cumulative splits plus demographics. Light Gradient-Boosting Machine also receives the extended pacing and latent features from Section IV-F. This gives the nonlinear model every advantage.

ΔRMSE is Ridge RMSE minus Light Gradient-Boosting Machine RMSE. Positive values mean Light Gradient-Boosting Machine has lower error. Negative values mean Ridge regression has lower error. The confidence intervals are conservative. They are built from independent bootstrap bounds because the two models use different conformal calibration procedures (LOO for Ridge regression, split-conformal for Light Gradient-Boosting Machine). Their bootstrap samples are not naturally paired.

Checkpoint	Ridge	LGBM	ΔRMSE	95% CI	Ridge $2\hat{q}$	LGBM $2\hat{q}$
5K	1250.8	1194.8	56.1	[18.2, 94.1]	54.6	51.4
40K	123.0	132.9	−10.0	[−69.0, 49.2]	2.8	3.1

TABLE VIII: Ridge (linear) vs. LightGBM (nonlinear). ΔRMSE = Ridge − LightGBM (positive favors LightGBM). Conservative BCa 95% CIs from independent bootstrap bounds. $2\hat{q}$ in minutes.

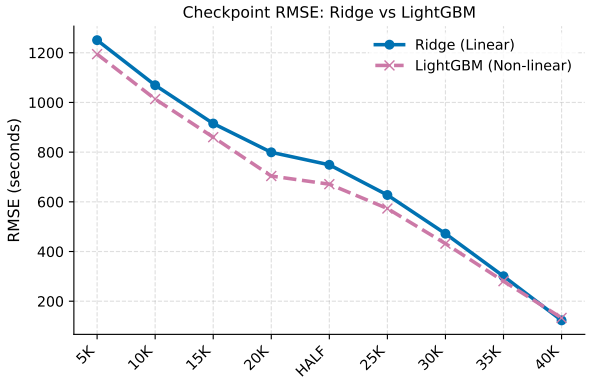


Fig. 10: RMSE by checkpoint for ridge and LightGBM. The gap narrows across checkpoints and vanishes by 40K.

At 5K, Light Gradient-Boosting Machine reduces RMSE by 56.1 seconds [18.2, 94.1]. The confidence interval excludes zero, so the improvement is statistically significant. But this gain is an order of magnitude smaller than the 876-second improvement from adding the 5K split itself. Information, not model complexity, is the main driver of accuracy at this checkpoint.

The Light Gradient-Boosting Machine advantage shrinks steadily across checkpoints (Figure 10). At 40K, the difference reverses. Ridge achieves RMSE of 123.0 versus 132.9 for Light Gradient-Boosting Machine. The difference is −10.0 seconds [−69.0, 49.2], which is not statistically significant. The conformal intervals tell the same story. Light Gradient-Boosting Machine's interval (3.1 minutes) is wider than Ridge regression's (2.8 minutes) at 40K. The extra flexibility of gradient boosting adds variance without reducing bias.

The small Light Gradient-Boosting Machine advantage at 5K likely reflects real nonlinearity in how early pace maps to final time. For example, runners who start faster than they can sustain may fade more than expected. Tree-based models can capture this pattern. Linear models cannot. As more checkpoints arrive, the model directly observes the runner's deceleration. The nonlinear structure that Light Gradient-Boosting Machine exploits gets absorbed into the data itself.

This is consistent with an information frontier. When the cumulative split vector captures the relevant predictive structure, a more flexible model has nothing left to exploit. Prediction accuracy here is governed by the information available, not by model complexity.

VII. Discussion

The central finding is that the information frontier governs prediction accuracy at every stage. This extends to any sequential prediction setting where covariates accumulate over time.

a) Switching criterion.

The conformal crossover gives a practical model-selection rule. A prediction system monitors conformal interval width. It switches from pre-race to in-race prediction at the first checkpoint where the in-race interval is narrower than the pre-race model's. In our data, this crossover occurs at 5K, the earliest available checkpoint. The same idea applies in other domains. For example, clinical deterioration models can switch from intake-based to lab-based prediction when the lab-based conformal interval drops below the intake model's. No domain-specific threshold tuning is needed.

b) Feature decomposition in sequential prediction.

The matched-λ decay analysis isolates each feature group's marginal contribution by fixing the regularization parameter across feature sets. This eliminates confounding between feature value and regularization strength. The design works wherever covariates arrive

in stages. Examples include demand forecasting as intraday sales accumulate, or credit scoring as transaction history grows.

c) Distributional shift.

The 2017 test year was 18.9°F above the training mean. This makes 2017 a conservative test case. The RMSE reductions and hierarchy ordering are preserved despite the covariate mismatch. However, empirical coverage at in-race checkpoints falls 4 percentage points below the nominal 90% target. This gap is consistent with a violation of the exchangeability assumption caused by the temperature shift. Adaptive conformal methods that condition on race-day covariates [13] could restore nominal coverage by widening intervals under adverse conditions.

d) Pacing structure.

The near-universal positive-split pattern (93%) constrains model design. Second-half deceleration is the main source of within-race variability. This explains why the demographic contribution peaks at 15K: age and sex predict fatigue patterns that early splits do not yet capture. The pacing distribution is similar across genders (Figure 3). This is consistent with our use of interaction terms rather than gender-stratified models.

VIII. Limitations

All evaluation uses a single test year (2017) from a single course. The Boston Marathon has a point-to-point profile and requires qualifying times for entry. This produces a nonrepresentative runner population. The information hierarchy may differ for open-entry or loop-course races.

Runner identity is inferred from display names using an automated collision-detection heuristic based on age-year consistency. Misattribution for common names could affect the mixed-effects model and the cluster-bootstrap. Both rely on correct runner-level grouping.

No physiological or training-load data is available. The information decomposition is limited to registration-time demographics, prior results, and race-day observables. Richer pre-race features (such as training volume or VO_2max) could narrow the gap between pre-race and in-race prediction. This could shift the crossover point to a later checkpoint.

Temperature is not modeled as an explicit predictor. The in-race models absorb heat effects indirectly through slower splits. Conditioning on race-day weather could improve pre-race predictions and reduce the coverage gap under distributional shift.

Light Gradient-Boosting Machine hyperparameters are fixed by design. This bounds the nonlinear gains from below. However, it does not invalidate the complexity-benchmark conclusion. The model-family effect (56 seconds at 5K) is an order of magnitude smaller than the information effect (876 seconds). Residual tuning gains cannot close this gap.

IX. Conclusion

Marathon finish-time prediction accuracy is governed by information availability, not model complexity. A nested five-model ablation with BCa cluster-bootstrap confidence intervals quantifies the marginal contribution of each information source. The first 5K split reduces RMSE by 876 seconds, more than doubling the combined gain from all pre-race features. Conformal interval widths identify when each source stops improving prediction. The personalization crossover, the checkpoint at which a Ridge regression model with one split produces a narrower interval than a personalized mixed-effects model, offers a practical switching criterion. The complexity benchmark confirms that gradient-boosted trees and Ridge regression converge to the same accuracy by 40K. These results establish the information frontier as the binding constraint on prediction accuracy.

Contributions

1) **H. Bubalo** implemented the Ridge regression checkpoint models with generalized cross-validation, the LightGBM gradient-boosted tree models with extended pacing and latent features, designed the model-complexity benchmark comparing linear and nonlinear model families and contributed to the writing of the paper.
2) **I. Roy** implemented the OLS models with demographic and history covariates including log-scale bias-correction retransformation, the linear mixed-effects model with runner-specific random intercepts and age slopes, the variance decomposition and shrinkage analysis, the conformal prediction framework for distribution-free prediction intervals, and contributed to the research and writing of the paper.
3) **J. Waga** implemented the bias-corrected and accelerated (BCa) cluster-bootstrap confidence

intervals with runner-level resampling and contributed to the research and writing of the paper.

4) **M. Tobe** conducted the pacing strategy analysis, including the classification of negative, even, and positive splits, the finish-time distributional comparisons across pacing groups, and the gender-stratified pacing breakdown and contributed to the research and writing of the paper.
5) **S. Bradley** conducted the race-day temperature analysis, including the Spearman rank correlation between maximum temperature and mean finish time, the per-age-group linear regression slopes, and the Mann–Whitney U warm-versus-cool significance tests across demographic subgroups and contributed to the research and writing of the paper.

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